

Unit Outline

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SWE30003

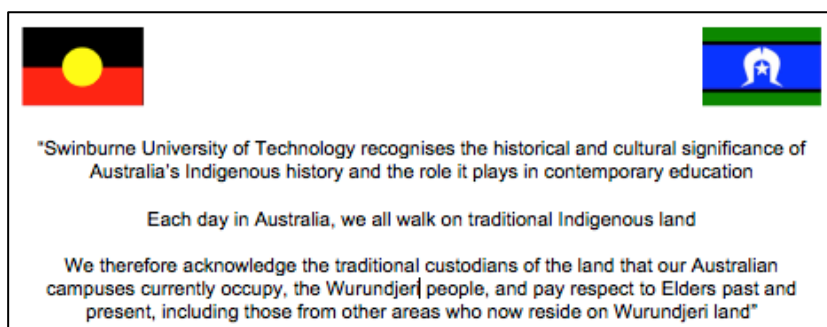
Software Architectures and Design

Semester May, 2026

Note that the lectures are to be conducted live online via Canvas/Collaborate-Ultra, and tutorial classes will be conducted face-to-face on campus.

Please read this Unit Outline carefully. It includes:

- PART A** Unit summary
- PART B** Your Unit in more detail
- PART C** Further information



PART A: Unit Summary

Unit Code(s)	SWE30003
Unit Title	Software Architectures and Design
Duration	One semester or 12 weeks
Total Contact Hours	36 hours
Requisites:	
Pre-requisites	COS20007 Object Oriented Programming AND 150 Credit Points
Co-requisites	
Concurrent pre-requisites	
Anti-requisites	
Assumed knowledge	
Credit Points	12.5
Campus/Location	Hanoi - HCM - Danang
Mode of Delivery	Lectures and Tutorials
Assessment Summary	Weekly questions and answers submissions, three group-based assignments, and final assessment (test / exam). As the minimum requirements of assessment to pass the unit and meet all Unit Learning Outcomes to a minimum standard, a student must achieve: (i) an aggregate mark of 50% or more, and (ii) at least 40% in the final assessment test, and (iii) at least 40% of the possible marks for the weekly questions and answers. Students who do not successfully achieve hurdle requirements (ii) and (iii) will receive a total mark for the unit of at most 45%.

Aims

The unit aims to facilitate an in-depth study of state-of-the-art approaches and techniques for system design with a special focus on the relationship between non-functional requirements and software architectures.

Unit Learning Outcomes

Students who successfully complete this unit can:

ULO1	Given a set of user goals and priorities, identify, formulate and analyse the requirements and constraints of a software system under consideration (K3, K6, S1, A2, A4, A5)
ULO2	Identify, construct and justify design abstractions at suitable levels for given user goals and system constraints (K3, K6, S1, S2, S3, A4, A6)
ULO3	Analyse a given system design in terms of the common design patterns used, and its benefits, limitations and appropriateness for the given user goals and priorities (K3, S1, A2, A4)
ULO4	Given a system's specification, formulate and document architectural decisions in terms of common patterns, and analyse the benefits and limitations of these decisions (K3, S1, S3, A4, A5)
ULO5	Record and describe their design decisions and structures using a contemporary modelling language (K3, S1, S3, A4, A5)

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit contributes to the development of the following Swinburne Graduate Attributes:

- GA2 Communication 2 - Communicating using different media
- GA3 Teamwork 1 - Collaboration and negotiation
- GA5 Digital literacies 1 – Information literacy
- GA6 Digital Literacies 2 – Technical literacy

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

Specifications

- User goals
- Constraints
- Priorities
- Validation
- Analysis

Abstractions and Patterns

- Abstractions in software
- Levels of abstraction
- Object-oriented abstractions
- Common patterns (structural and behavioural)
- Pattern recognition and identification
- Pattern systems

Software Architectures

- Decisions vs. structures
- Architectural patterns
- Documentation and communication

Architectural Styles

- Message-driven architectures
- Client-server architectures
- Layered Architectures

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- Making all assignments groupwork,
- Clarifying the expectations and their differences from other/prior units, and
- Introducing new architecture styles, including service-oriented architecture (SOA).

Unit Teaching Staff

Name	Role	Email	Consultation Times
Mr. Duc Le	Unit Convenor in Hanoi	dle1@swin.edu.au	See Canvas
Mr. Khanh Dang	Unit Convenor in HCM	ktdang@swin.edu.au	See Canvas
Mr. Tuan Tran	Unit Convenor in HCM	tuanatran@swin.edu.au	See Canvas
Mr. Hoang Nhu Vinh	Unit Convenor in Danang	nhoang@swin.edu.au	See Canvas

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
Live Online	Lectures	24 hours	2 hours	Weeks 1 to 12
In person	Class (Tutorials)	12 hours	1 hour	Weeks 1 to 12
Online	Directed Online Learning and Independent Learning	24 hours	2 hours	Weeks 1 to 12
Unspecified Activities	Independent Learning	90 hours	7.5 hour	Weeks 1 to 12

Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	May 11	<i>Lecture:</i> Introduction, Issues in Software Design <i>Tutorial:</i> Introduction and Group Formation	Assignment Group Formation
2	May 18	<i>Lecture:</i> Goal-Design Scale, User Tasks <i>Tutorial:</i> Software Design Issues	Assignment Group Formation due
3	May 25	<i>Lecture:</i> Quality Attributes, Requirements Validation <i>Tutorial:</i> User Tasks	

4	Jun 01	<i>Lecture:</i> Domain Analysis, Domain Modelling, Software Abstractions <i>Tutorial:</i> Quality Attributes; Assignment 1 Q&A	
5	Jun 08	<i>Lecture:</i> Responsibility-Driven Design / Object Oriented Design <i>Tutorial:</i> Requirements Analysis Exercise	
6	Jun 15	<i>Lecture:</i> Detailed Object Design <i>Tutorial:</i> Object Oriented Design	Assignment 1 (Requirements Specification) due
7	Jun 22	<i>Lecture:</i> Case Study in Object Design <i>Tutorial:</i> OO Design Exercises (II part 1); Assignment 2 Q&A	
8	Jun 29	<i>Lecture:</i> Design Patterns <i>Tutorial:</i> Object-Oriented Design (II part 2), Assignment 1 feedback	
9	July 13	<i>Lecture:</i> Software Architectures, Architectural Styles <i>Tutorial:</i> Software Architecture and Patterns	Assignment 2 (Object Design I) due
10	July 20	<i>Lecture:</i> Case Study in Architectural Design <i>Tutorial:</i> Architectural Design (I); Assignment 3 Q&A	
11	July 27	<i>Lecture:</i> Documenting Designs <i>Tutorial:</i> Architectural Design (II); Assignment 2 Feedback	
12	Aug 03	<i>Lecture:</i> Summary of Main Concepts, Wrapping Up <i>Tutorial:</i> Architecture Tactics, Documentation, and Evaluation; Q&A	Assignment 3 (Object Design II) due
13			Final Assessment (in-person, invigilated)

Assessment

a) Assessment Overview

Tasks and Details	Individual or Group	Weighting	Unit Learning Outcomes that this assessment task relates to	Assessment Due Date
Weekly Questions and Answers	Individual	10%	All	Weeks 2 – 11
Requirements Specification	Group	15%	1	Week 5

Object-Oriented Design (Part I)	Group	20%	2, 3, 4, 5	Week 9
Object-Oriented Design (Part II)	Group	15%	2, 3, 4, 5	Week 12
Final Assessment (test/exam)	Individual	40%	All	Formal Assessment Period

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	The assessments in this unit should be completed entirely WITHOUT genAI assistance. genAI may be used for self-study and research purposes. However, be warned that answers from genAI may not be correct or accurate.
c) Hurdle requirements	As the minimum requirements of assessment to pass this unit and meet all ULOs to a minimum standard, a student must have achieved: (i) an aggregate mark for the unit of 50% or more, (ii) at least 40% in the final assessment, and (iii) at least 40% of the available marks for the Weekly Questions and Answers submissions. Students who do not successfully achieve hurdle requirements (ii) and (iii) of the above requirements, will receive a total mark for the unit of at most 45%.
d) Final assessment period	If the unit you are enrolled in has a final assessment (including invigilated exams), you will be expected to be available for the entire final assessment period including any Special Exam period. For this unit, a final assessment is to be conducted as an in-person invigilated test/exam during the final assessment/exam period.
e) Submission requirements	Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin . Please ensure you keep a copy of all assessments that are submitted. An Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Submitting work webpage or www.swinburne.edu.au/studentforms/

	The exact deliverables for each assignment will be outlined in the corresponding assignment specification. In general, the deliverables are to be submitted through Canvas or other means to be specified in the assignment specifications. The Weekly Question and Answer submission will be in electronic form, and submitted through Canvas. For every group assignment, a contribution sheet <i>must be signed by all group members</i>, and submitted with the assignment. <i>Group assignments submitted without the signed contribution list may not be marked and may not receive any feedback.</i> Individual members and the group must include a statement of evidence to substantiate their individual contributions and collaborations, as part of the contribution sheet.
f) Extensions and late submissions	Late Submissions - Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 10% of your achieved mark for each working day the task is late, up to a maximum of 5 working days. After 5 working days, a zero result will be recorded. Extensions for ongoing assessments are available for medical reasons (doctors certificate must be provided). Students must apply for an extension by emailing the Unit convenor <i>at least 48 hours prior to the due date</i> (unless in emergency situations) and also must supply any supporting documentation if requested. In general, no late submission will be granted for the Weekly Question and Answer submissions.
g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are: Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/
h) Groupwork guidelines	A group assignment is the collective responsibility of the entire group, and if one member is temporarily unable to contribute, the group should be able to reallocate responsibilities to keep to schedule. In the event of longer-term illness or other serious problems involving a member of group, it is the responsibility of the other

	members to notify immediately the Unit Convenor and relevant tutor. Group submissions must be submitted with an Assignment Cover Sheet, signed by all members of the group. All group members must be satisfied that the work has been correctly submitted. Any penalties for late submission will generally apply to all group members, not just the person who submitted.
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Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

Here is a list of recommended reading materials for this unit:

- Soren Lauesen, *Software Requirements: Styles and Techniques*, Addison-Wesley, 2002
- Len Bass, Paul Clements, and Rick Kazman, *Software Architecture in Practice* (4th Edition), Addison-Wesley, 2021
Please note that the 3rd or 2nd edition are also available through the Swinburne library and can be referenced..
- David Budgen, *Software Design* (2nd Edition), Addison-Wesley, 2003
- Eric Evans, *Domain-Driven Design*, Addison-Wesley, 2004
- Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides, *Design Patterns*, Addison-Wesley, 1995
- Frank Buschmann, Regine Meunier, Hans Rohnert, Peter Sommerlad and Michael Stal, *Pattern-Oriented Software Architecture: A System of Patterns*, Wiley, May 1996
- Ian Sommerville, *Software Engineering* (8th Edition), Addison-Wesley, August 2007
- Craig Larman, *Applying UML and Patterns* (3rd Edition), Prentice Hall, 2005
- Rebecca Wirfs-Brock and Alan McKean, *Object Design*, Addison-Wesley, 2003
- Jeff Garland and Richard Anthony, *Large-Scale Software Architecture*, Wiley, 2003.

Further reading material will be provided during the lectures and/or made available on Canvas.

PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage <http://www.swinburne.edu.au/student/>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/> Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence;

contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non-compliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see <https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the [Current students](#) web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See <https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/>

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact [Accessability Services](#) to discuss further.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

If you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Independent Advocacy Officer at the Swinburne Student Association. Talking to an Advocacy Officer is free, independent and confidential. For more information and booking an appointment, please see <https://www.swinburne.edu.au/current-students/student-services-support/advocacy/>